

Five Lessons from the Mammogram Debate that Connect to Block 1

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1. A positive test usually does *not* mean cancer (base rates)

Suppose that among women in their 40s who get screened, about 1% have breast cancer; the current screening/test catches about 90% of true cancers but also flags about 9% of healthy women (9% false-positive rate). **Question: a woman tests positive — what is the chance she has cancer?** Most people (including some physicians) say 80–90%. Count it out instead. Out of 1,000 women screened: 10 have cancer, and 9 of them test positive. Of the 990 healthy women, about 89 *also* test positive. So 98 women get a scary phone call, and only 9 of them have cancer — **about 1 in 10**. When the condition is rare, most positives are false positives, no matter how good the test is. *Business version:* fraud alerts, resume screens, quality inspections — any “detector” aimed at a rare event.

2. “Cuts deaths by 20%” and “saves 1 life per 1,000+ screened” are the same fact

Screening women in their 40s reduces breast cancer mortality by roughly 15–20% (*relative risk*). But because dying of breast cancer in one’s 40s is uncommon, that translates to roughly one death averted per 1,000+ women screened for a decade (*absolute risk*) — alongside hundreds of false alarms. Relative risk makes any effect sound big; absolute risk tells you what it means for one person. **Always ask: “percent of what?”**

3. Better survival $\neq$ fewer deaths (lead-time bias)

Screening finds cancers earlier, which starts the “survival clock” earlier. Five-year survival can improve *even if no one lives a single day longer* — patients just spend more of their remaining time counted as “survivors.” Screening also preferentially catches slow-growing tumors (fast ones appear between screenings). This is why serious evaluations score screening on **death rates, never survival rates**. *Business version:* any metric where measurement changes when a case enters the data — e.g., “time-to-resolution fell” after you started logging tickets earlier (a form of survival bias)

4. Finding more cancer isn’t automatically good (overdiagnosis)

Some screen-detected tumors would never have caused harm — but every one found gets treated, and every treated patient understandably credits screening with saving their life. Individually sincere testimonials can add up to a misleading aggregate picture. More detection is not the same as more lives saved.

5. Guidelines flip-flop because the *trade-off* is close, not because the science is shaky

Official advice has swung: start at 50 (2009, 2016), start at 40 (2024). Meanwhile the American Cancer Society says 45 and radiologists say every year from 40. The underlying trials barely changed — what differs is how each body *weighs* the same numbers: one averted death vs. hundreds of false positives and some overdiagnosis. **Same data, different loss functions lead to different decision.**

6. What is the Population of Interest?

The RCTs behind the mortality estimates are mostly from the 1960s-90s, when treatment was far worse — so the “population of interest” (women screened in 2026 under modern therapy) differs from the trial populations along dimensions no reweighting fixes, since the treatment-era interaction isn’t an observable you can weight on or match.

Takeaway: before trusting any performance claim, ask four questions. What’s the base rate? Percent of what? Did the measurement itself change? And whose weights turned this evidence into this recommendation?